HORN Preregistration Note: Directional Outlier Propagation in Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HORN tests whether attention-to-SSM and SSM-to-attention boundaries differ in outlier magnitude and quantization-noise sensitivity. The current Granite Tiny H1a and H2 scouts are negative, so no positive result or native systems claim is made.

1 Current Gate

The branch must pass activation-magnitude, quantization-noise propagation, and cross-model/control gates before GPU validation. H1a first screens one live hybrid model for directional activation statistics, H1 requires the same selected direction on at least two hybrid models, H2 tests matched FP4-equivalent noise propagation, and H3 requires the direction to survive hybrid-model repeats while weakening on pure-attention and pure-Mamba controls. The current real trace packet builder/checker covers H1a/H1 boundary statistics. H2 and H3 now have executable follow-up contracts in `experimental/shared/followup_gate_contracts.py`: they require a fixed H1-selected direction, exact noise side and noise scale basis, paired clean/noisy rows, hook-off controls, and pure-attention plus pure-Mamba control rows. The current resource-limited H2 scout satisfies the H2 contract but fails the gate. No noise-propagation, precision-allocation, or cross-architecture claim is allowed unless the branch is reopened with a fuller preregistered H2/H3 scope.

2 Mac Artifact Status

A deterministic synthetic H1a packet validates the directional-boundary real-schema contract. It emits 72 rows, selected max-abs ratio 4.044, non-boundary selected-direction control ratio 1.042, and permuted selected-direction control ratio 0.247, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_HORN_H1A`. This is synthetic-only: it is not model evidence, not a quantization-noise result, and not GPU evidence. The current Granite Tiny H1a packet uses 12 real prompts over all 8 planned boundaries with right-layer input hooks and passes the real checker in resource-limited non-promoting mode, but its selected ratio is only 1.06. The current H2 scout uses 2 prompts, 3 seeds, paired boundary-direction noisy continuations, and hook-off controls. It is contract-valid with paired units 6/6 and hook-off max delta 0.0, but its directional drift ratio is only 1.037, signed selected-direction lower bound is 0.324, and selected-direction support is 0.5, producing `FAIL_REAL_HORN_H2_DIRECTIONAL_NOISE_PROPAGATION`. This demotes HORN as a standalone branch.

3 Next Evidence

The next admissible action is not GPU validation. HORN remains negative/control appendix evidence unless a future full H2/H3 reopening is preregistered with more prompts, more models, and a clear reason why the current near-null H2 scout should be overturned. Pure-architecture controls remain an H3 requirement for any future revival. The paper is not camera-ready as a method or measurement paper until real H1–H3 evidence exists; more precisely, the current blocker is passing H1–H3 evidence rather than the existing negative scouts.

4 Admissible Real Packet

The real H1a/H1 packet must use the shared config-derived architecture maps rather than substring-only module classification. Each row must report `model_id`, `prompt_id`, adjacent layer IDs, `direction`, `matched_boundary_direction`, `boundary_index`, normalization placement, `max_abs`, `rms`, `kurtosis`, and `control_type`. The required controls are boundary rows, non-boundary adjacent rows, and permuted-direction rows. A valid packet must cover both `attention->ssm` and `ssm->attention` in boundary rows and in matched control labels. The frozen trace plan schedules a matched non-boundary control for each observed boundary row, not merely one aggregate control per direction. Non-boundary controls may retain their true architecture direction, such as `ssm->ssm`, while `matched_boundary_direction` records the boundary direction they control for, and both matched directions must appear for every prompt; each permuted-direction row must match an observed boundary tuple with the same prompt ID, layer IDs, normalization positions, and measured metrics while flipping the actual `direction` label only. It cannot substitute a separately measured tensor for the null: the control is built by reusing the observed metrics and flipping its direction label. It cannot pass if the permuted controls preserve the selected high-magnitude direction; a faithful label flip may preserve unsigned asymmetry while moving the signal to the opposite label. Non-boundary controls must stay below the selected H1 threshold, not merely below the observed boundary ratio. It must also include at least 12 fixed prompt IDs unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The packet must include `prompt_ids_hash`, `architecture_map_hash`, `trace_plan_hash`, per-row `prompt_cluster_id`, the exact `trace_plan_path` whose file SHA-256 matches that hash, and a summary with prompt count, boundary directions, selected H1a ratio, selected H1a prompt-cluster-bootstrap lower bound, support fraction, non-boundary control ratio, and permuted-direction ratio, then pass `experimental.shared.check_gate_packet --mode real --project horn`. The checker recomputes these H1a fields from raw rows and rejects stale or hand-filled summaries. Saved tensor packets must be converted with `experimental.shared.hybrid_trace_packet_builder` before validation. The current capture-manifest templates are generated by `experimental.shared.hybrid_trace_capture_manifest`; they must be filled from real boundary-activation captures before packet building. Served HF IDs are preserved as `served_model_id` while rows are validated against the canonical architecture-map `model_id`; permuted-direction entries use `tensor_alias_of` to reuse the observed boundary tensor. Saved tensor packets must carry `tensors/tensor_manifest.json` and the referenced tensor files. Each raw row copies `tensor_name`, `tensor_source_name`, `tensor_storage_name`, `tensor_sha256`, `tensor_dtype`, and `tensor_shape` from that manifest; permuted rows must reuse the observed boundary source and hash rather than introduce independently measured tensors.

Required control	Reviewer objection addressed
Explicit boundary map	Avoids substring artifacts.
Non-boundary adjacent pair	Separates boundary effects from depth effects.
Permuted direction label	Tests whether direction carries signal.
Matched normalization side	Controls RMSNorm/LayerNorm placement.
Pure-architecture control	Prevents generic outlier claims.

Table 1: Controls required before HORN can claim directional hybrid asymmetry.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/horn/paper/reviewer_pack.md`. It records that HORN is not camera-ready as a standalone method or measurement paper and should currently serve only as negative/control evidence. No native systems claim is made yet.